Project M-087 Troubleshooting Record

1 Summary

Prototype M-087 exhibited excessive vibration at operating speed.

2 Investigation

Bearing alignment and mounting were within tolerance.

3 Root Cause

Dynamic balancing showed an imbalanced rotor, 14 g-mm residual imbalance, exceeding the G2.5 limit. Excessive vibration was associated with the imbalanced rotor.

4 Fix

Rotor rebalanced by material removal; vibration dropped below 2 mm/s RMS.